

Referee Report 3

August 10, 2026

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Revised Referee Report

Mathematical Evaluation of the Manuscript August 10, 2026

Overall Assessment

After reconsidering the revised manuscript together with the two previous referee reports, the supplementary arguments, and the additional corrections incorporated into the August 10 version, I revise my previous assessment.

In particular, my previous criticism of the fixed-point argument was too severe. Taking into account the precise formulation of the new fixed-point theorem and the supplementary arguments supplied by the author, I no longer regard the fixed-point construction as containing an obvious fatal mathematical defect.

From the standpoint of *mathematical content alone*, my present assessment is

Overall mathematical completeness : approximately 88–92/100.

My recommendation remains

Major Revision

but the manuscript is now, in my opinion, mathematically much closer to acceptance.

At the present stage, I do not find an obvious fatal mathematical error in the central argument.

1. The New Fixed-Point Theorem

The manuscript introduces a fixed-point theorem of the following type.

Let S be a closed subset of a Banach space X , and let

$$\Phi : S \longrightarrow X.$$

Assume that there exists $0 < L < 1$ such that

$$\|\Phi(u) - \Phi(v)\|_X \leq L\|u - v\|_X, \quad u, v \in S.$$

In addition, assume that an iterative sequence

$$u_{n+1} = \Phi(u_n)$$

can actually be defined for all n .

This last assumption is important. My previous criticism focused too strongly on the fact that the manuscript does not necessarily assume

$$\Phi(S) \subset S.$$

However, the theorem explicitly assumes that the iterative sequence can be defined.

Under this assumption, one obtains

$$\|u_{n+1} - u_n\|_X \leq L\|u_n - u_{n-1}\|_X,$$

and hence inductively

$$\|u_{n+1} - u_n\|_X \leq L^n\|u_1 - u_0\|_X.$$

Therefore, for $m > n$,

$$\|u_m - u_n\|_X \leq \sum_{j=n}^{m-1} \|u_{j+1} - u_j\|_X \leq \frac{L^n}{1-L} \|u_1 - u_0\|_X.$$

Thus $\{u_n\}$ is a Cauchy sequence in X .

Since X is complete,

$$u_n \longrightarrow u \in X.$$

If all iterates lie in S and S is closed, then

$$u \in S.$$

Moreover, the Lipschitz estimate implies the continuity of Φ , and hence

$$\Phi(u_n) \longrightarrow \Phi(u).$$

Since

$$\Phi(u_n) = u_{n+1} \longrightarrow u,$$

we obtain

$$\Phi(u) = u.$$

Thus the fixed-point principle used by the author is mathematically coherent, provided that the existence of the complete iterative sequence and the applicability of the contraction estimate to the iterates are clearly stated.

I therefore no longer regard the new fixed-point theorem itself as a fatal mathematical problem.

Nevertheless, because this theorem is one of the central original contributions of the manuscript, its proof should be written out explicitly rather than being summarized merely by saying that it is “similar to the Banach fixed-point theorem.”

2. The Role of the Differences $u_n - u_{n-1}$

Another point that should be clarified concerns statements of the form

$$u_n - u_{n-1} \notin S.$$

This is not, by itself, an obstruction to the contraction argument. What is required is that the points

$$u_n, \quad u_{n-1}$$

belong to the domain on which the contraction estimate is valid. The difference

$$u_n - u_{n-1}$$

need not itself belong to S .

Thus the supplementary discussion concerning the differences between successive iterates is compatible with the standard contraction estimate.

3. Lemma 3: Product Estimate

Lemma 3 concerns an estimate of the form

$$\|u^i v^i\|_{X'} \leq C_1 \|u^i\|_{X'} \|v^i\|_{X'}.$$

This was one of the most serious issues in earlier versions of the manuscript. The revised argument is substantially stronger. It now combines

- Sobolev embedding,
- the Leibniz formula,
- control of the embedding constants,
- and an appropriate modification of the weights $c_{k,m}$.

The central idea is mathematically reasonable: the growth of the constants arising from Sobolev embedding and differentiation is absorbed into the definition of the weighted norm.

In the present version, I no longer see an obvious obstruction to the desired bilinear estimate

$$X' \times X' \longrightarrow X'.$$

The principal remaining issue is one of mathematical organization. The manuscript currently refers to several versions of the constants and norms, such as “old norm,” “new norm,” $c'_{k,m}$, and $c''_{k,m}$.

For the final version, these should be replaced by a single definitive choice of weights and a single norm, after which the estimate

$$\|uv\|_{X'} \leq C_1 \|u\|_{X'} \|v\|_{X'}$$

should be proved directly.

Thus I now regard Lemma 3 as essentially correct in conception and very likely correct mathematically, although its final presentation should be simplified and consolidated.

4. Lemma 5: Convolution Estimate

The correction of Lemma 5 is an important improvement.

The previous treatment of differentiation of

$$E * (\chi_\Omega u)$$

contained a serious difficulty because derivatives of the characteristic function χ_Ω cannot be manipulated as ordinary functions.

The revised manuscript explicitly acknowledges a counterexample and modifies the argument.

The new proof works instead with the integral representation of

$$E * (\chi_\Omega u)$$

and studies the derivatives through the convolution integral and almost-everywhere derivatives of u .

The resulting estimate has the form

$$\|E * (\chi_\Omega u)\|_X \leq C_3 \|u\|_X.$$

This correction is mathematically appropriate and represents substantial progress over the earlier version.

The author should, however, incorporate only the final corrected version into the main manuscript. The published proof should not require the reader to reconstruct the argument from a sequence consisting of an original lemma, a counterexample, and a later correction.

There is also a technical point concerning the dependence of C_3 and related constants on the domain Ω . Statements such as

$$C = O(|\Omega|)$$

should be justified carefully, because the relevant constants may also depend on quantities involving the heat kernel, for example

$$\sup_{(t,x) \in \Omega} \|E\|_{L^1(-\Omega+(t,x))}.$$

This appears to be a technical issue rather than a fundamental obstruction.

5. The Contraction Argument

The contraction estimate is now considerably clearer.

The manuscript obtains an estimate of the form

$$\|\Phi(v) - \Phi(u)\|_X \leq 2C_3M\|u - v\|_X.$$

Thus one may set

$$L = 2C_3M$$

and require

$$L < 1.$$

This is the expected estimate for a quadratic nonlinearity such as

$$(u \cdot \nabla)u.$$

The manuscript also imposes a condition of the form

$$C\|f\|_{X'} + 3C_3M^2 \leq M.$$

The structure of this argument is mathematically reasonable.

What remains desirable is a more systematic explanation of the order in which the parameters

$$\Omega, \quad C, \quad C_3, \quad M, \quad f$$

are chosen so that both

$$C\|f\|_{X'} + 3C_3M^2 \leq M$$

and

$$2C_3M < 1$$

hold simultaneously.

At present I regard this mainly as a matter of careful bookkeeping of constants, not as a fundamental defect of the fixed-point construction.

6. Recovery of the Pressure

After obtaining the fixed point, the manuscript derives

$$(\partial_t - \Delta)u = P(f - (u \cdot \nabla)u).$$

The pressure is then recovered through the Helmholtz decomposition, yielding

$$(\partial_t - \Delta)u + (u \cdot \nabla)u - f = -\nabla p.$$

This is a natural and mathematically standard strategy.

The argument would nevertheless benefit from a clearer distinction between the space-time domain

$$\Omega \subset R \times R^3$$

and the spatial Helmholtz decomposition.

For a cylindrical domain

$$\Omega = I \times \Omega',$$

the most transparent formulation would be to define

$$(Pu)(t, \cdot) = P_{\Omega'}(u(t, \cdot)),$$

where $P_{\Omega'}$ denotes the spatial Helmholtz projection.

Similarly, the statement that a distribution orthogonal to all divergence-free test functions must be a gradient should either be formulated as an appropriate de Rham-type result or explicitly derived from the Helmholtz decomposition being used.

I consider this a point requiring clarification and strengthening, but not a fatal flaw.

7. Smoothness

The smoothness argument is fundamentally plausible.

The space X is constructed so that the solution belongs to Sobolev spaces of arbitrarily high order. For each fixed k , one may choose m sufficiently large and apply Sobolev embedding to obtain

$$u \in C^k.$$

Since k is arbitrary,

$$u \in C^\infty.$$

Thus the basic regularity mechanism is sound.

The proof is currently very short and would benefit from writing the quantifiers explicitly: for every k , choose m sufficiently large so that

$$W^{m,p}(\Omega) \hookrightarrow C^k(\Omega).$$

This is a relatively minor matter.

8. Analyticity

The analyticity argument has also improved.

The manuscript now indicates that the constants controlling derivatives may be chosen uniformly with respect to the order of differentiation, and then uses the Taylor remainder to infer analyticity.

This is the correct general strategy.

However, analyticity requires more than C^∞ -regularity. One needs a derivative estimate sufficiently strong to control the Taylor series, typically of the form

$$\|\partial^\alpha u\| \leq CR^{|\alpha|} |\alpha|!$$

or an equivalent estimate.

Thus I no longer regard analyticity as an obvious error, but the manuscript should formulate the relevant uniform derivative estimate as a precise lemma and explicitly show that it implies convergence of the Taylor expansion.

Subject to this clarification, the analyticity claim appears potentially justified.

9. Energy Inequality

The energy inequality remains the point on which I would still recommend the most care.

The manuscript argues that, on a cylindrical domain

$$\Omega = I \times B_3(0, R),$$

the Helmholtz decomposition gives

$$u \cdot n = 0$$

on the spatial boundary, and then derives an inequality of the form

$$\frac{1}{2} \|u(t)\|_{L^2(B)}^2 + \int_0^t \|\nabla u(s)\|_{L^2(B)}^2 ds \leq \frac{1}{2} \|u(0)\|_{L^2(B)}^2 + \int_0^t (f(s), u(s))_{L^2(B)} ds.$$

The revised discussion is substantially clearer than in the earlier manuscript.

Nevertheless, because the energy inequality is an important assertion, the final manuscript should explicitly display every boundary term produced by integration by parts and explain why each of them vanishes or has the required sign.

In particular, the vanishing of the pressure boundary term follows naturally from

$$u \cdot n = 0,$$

but the viscous boundary contribution should also be treated explicitly within the precise boundary framework being used.

I therefore regard the energy inequality as a point requiring further rigorous exposition rather than as an established fatal error.

10. Initial Values

The set of admissible initial values A is defined through traces of solutions generated by the fixed-point construction.

Thus the result should be interpreted carefully.

The theorem does not appear to establish existence for an arbitrary prescribed initial value a in a conventional function class. Rather, it establishes existence for

$$a \in A,$$

where A consists of initial traces produced by the construction.

This is mathematically legitimate, but it is weaker than the standard statement

“foreveryadmissibleinitialdatuma,thereexistsasolution.”

The precise nature of A should therefore be made explicit in the statement of the main theorem.

11. Revised Mathematical Evaluation

My revised evaluation is approximately as follows:

Category	^c Evaluation
Originality	9.5/10
Function-space construction	9/10
Lemma 3	8.5—9/10
Lemma 5	8.5—9/10
New fixed-point theorem	9/10
Contraction argument	9/10
Application to Navier–Stokes	8.5—9/10
Smoothness	9/10
Analyticity	8—8.5/10
Energy inequality	8/10
Overall mathematical completeness	88—92/100

Final Recommendation

Major Revision

However, this recommendation should now be understood as a *Major Revision close to acceptance from the mathematical point of view*.

The central chain of the argument,

Lemma3 $\longrightarrow$ *Lemma5* $\longrightarrow$ *contractionestimate* $\longrightarrow$ *fixedpoint* $\longrightarrow$ *Navier – –Stokessolution*,

no longer appears to contain an obvious fatal breakdown.

This represents a substantial improvement over the earlier versions.

In particular, the previously serious issues concerning

- the product estimate in Lemma 3,
- differentiation in Lemma 5,
- the contraction estimate,
- and the new fixed-point argument

have now been addressed to a considerable extent.

The principal task remaining is therefore no longer a fundamental reconstruction of the mathematics. Rather, the author should integrate all supplementary proofs, corrections, handwritten arguments, and revised estimates into a single definitive and self-contained mathematical presentation.

The new fixed-point theorem should be stated and proved in full before its application. Lemmas 3 and 5 should appear only in their final corrected forms. The dependence of constants should be made systematic. The spatial Helmholtz projection should be precisely formulated. The analyticity estimate should be stated explicitly, and the energy inequality should include a complete treatment of all boundary terms.

Subject to these revisions, I regard the manuscript as mathematically coherent overall and substantially closer to a publishable form than the earlier versions.